Section-A

a) What do you mean by statistical data? What are the sources of statistical data? Distinguish between primary and secondary data.

b) What is central tendency? What are the measures of central tendency? Show that "Sum of the deviations from arithmetic mean is zero".

c) For two non-zero positive observations show that $AM \geq GM \geq HM$.

(a) What do you mean by dispersion? What are the important measures of dispersion? Define standard deviation and CV.

(b) If $\bar{x}$ and σ denote respectively the mean and standard deviation of a set of n observations, then show that $\bar{x}\sqrt{(n-1)} \geq \sigma$.

(c) Calculate the mean, standard deviation and CV from the given data: 20, 26, 40, 36, 23, 42, 35, 24, 30, 32.

(a) Define (i) Sample Space (ii) Random Experiment (iii) Probability of an Event.

(b) Three events A_1, A_2 and A_3 are mutually exclusive and their union is the sample space S. If $P(A_1) = \frac{3}{2}P(A_2), P(A_2) = 2P(A_3)$, then find $P(A_1), P(A_2)$ and $P(A_3)$.

(c) Write down the probability mass function of Poisson distribution. Find the mean and variance of Poisson distribution. Hence show that mean and variance of the distribution are equal.

Section-B

(a) What do you mean by scatter diagram? Define Pearson correlation coefficient. Mention the important properties of correlation coefficient.

(b) Show that correlation coefficient lies between -1 and +1. If X and Y are independent then show that they are uncorrelated.

(c) Calculate the correlation coefficient between temperature (x in K) and pressure (y in atm) of a certain gaseous product and interpret the result:

x	298	301	302	310	312	323	337	341	349
y	1.2	1.3	1.5	1.7	1.8	2.0	2.1	2.3	2.8

(a) What do you mean by a simple linear regression model? Distinguish between correlation and regression.

(b) Prove that the regression coefficient is independent of origin but not of scale.

(c) GPA (x) and the corresponding starting salary (y) in thousand were as follows:

x	3.26	2.60	3.35	2.86	3.82	2.21	3.47	3.28	2.54	3.25
y	33.8	29.8	33.5	30.4	36.4	27.6	35.3	35.0	26.5	33.8

Estimate the least squares prediction equation of y on x. Find the starting salary corresponding to each of the GPAs 2.75 and 3.75.

(a) Define (i) Type-I Error (ii) Type-II Error (iii) Level of Significance (iv) Power of a Test.

(b) A random sample of 15 men from a community yields a mean height of 67.6 inches. The value of standard deviation from the sample is 3.4 inches. Do the data support the hypothesis that the mean height of men in that community is 65.8 inches? $[t_{0.05,14} = 2.145]$

(c) Define 2×2 contingency table? For 2×2contingency table, show that

$$\chi^2 = \frac{N(ad - bc)^2}{(a+b)(c+d)(a+c)(b+d)}$$